Samira Khan

Electrical Engineer — Power Electronics, Controls & Embedded Systems

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1. Introduction / Professional Profile

I am an Electrical Engineer with 6 years of experience in power electronics, mixed-signal board design, and embedded firmware bring-up. I have taken designs from schematic and PCB layout through lab validation, EMI troubleshooting, and transfer to manufacturing.

I enjoy solving the interface between hardware and firmware: clean schematics, thoughtful layout, instrumented test plans, and documentation that manufacturing and field teams can trust.

2. Professional Summary

Experience with DC-DC converters, motor drive stages, sensor front-ends, STM32/ESP32 platforms, and lab tools (oscilloscope, logic analyzer, electronic load, spectrum analyzer basics). Strong collaborator with mechanical and software teams on system integration.

Seeking an Electrical Engineer position focused on product hardware, power systems, or embedded controls in a hands-on engineering culture.

3. Contact Information

- **Full Name:** Samira Khan
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- **Location:** Austin, TX, USA
- **LinkedIn:** linkedin.com/in/samirakhan-ee
- **Portfolio / Website:** samirakhan-ee.com
- **Availability:** Open to full-time opportunities

4. Education Details

B.S. in Electrical Engineering — The University of Texas at Austin

2014 – 2018 | GPA: 3.70 / 4.00 | Austin, TX

- Coursework: Power Electronics, Analog Circuits, Digital Logic, Control Systems, Embedded Systems
- Senior Design: Closed-loop BLDC motor controller with current sensing and GUI telemetry
- IEEE student branch volunteer for tutoring and lab safety workshops

Continuing Education — PCB Design for EMC — Industry short course

2021 | GPA: Completed | Austin / Online

- Covered return paths, stackups, filtering, and practical debug strategies for radiated emissions
- Applied learnings to reduce pre-compliance failures on a subsequent product spin

5. Core Skills

Technical Skills

- Circuit design: DC-DC (buck/boost), linear regulators, gate drivers, sensing (current/voltage/temperature)
- PCB: Altium Designer, KiCad; multilayer layout, controlled impedance awareness, DFM/DFT basics
- Embedded: C/C++ on STM32, ESP32; UART/SPI/I2C; FreeRTOS familiarity
- Controls: PID tuning basics, current/speed loops for small motor drives
- Validation: Bench bring-up, electronic load testing, thermal soak, EMI pre-compliance support
- Documentation: Schematics packages, BOMs, test procedures, ECO support

Tools & Platforms

- Oscilloscope, multimeter, logic analyzer, soldering/rework station, programmable PSU
- Altium, KiCad, MATLAB/Simulink basics, Git for firmware
- Jira, Confluence, manufacturer portals for component lifecycle checks

Soft Skills

- Clear lab notebooks and handoff notes for manufacturing test
- Cross-discipline integration with mechanical and firmware teammates
- Calm, structured debug under schedule pressure

6. Certifications & Training

- IPC-A-610 awareness training (electronics assembly quality)
- Internal ESD control and lab safety certification
- Vendor training: TI power design seminars (selected modules)

7. Additional Qualifications

Comfortable owning schematic-to-layout reviews and first-power-on checklists. Interested in roles involving battery-powered systems, industrial sensors, or renewable-adjacent power electronics.

Portfolio of non-confidential block diagrams and bring-up photos available on request.

8. Projects

48V-to-12V Isolated DC-DC Module

Electrical Design Owner | 2023 – 2024

Designed an isolated converter module for industrial sensing nodes with efficiency and thermal targets.

- Met >90% efficiency target at nominal load after two layout spins
- Defined protection features: OCP, OVP, and thermal shutdown coordination with firmware
- Wrote production test limits and partnered with manufacturing on ICT fixtures

BLDC Motor Drive Evaluation Board

Hardware + Firmware Bring-up | 2022

Three-phase inverter stage with current sensing and closed-loop speed control demo.

- Implemented FOC-lite control path with STM32 timers and ADC sampling
- Characterized torque ripple and improved PWM dead-time settings
- Created GUI telemetry for lab demos and customer evaluations

Low-Noise Sensor Front-End for Vibration Monitoring

Analog Design Engineer | 2021

Signal conditioning chain for piezoelectric sensors in predictive maintenance hardware.

- Improved SNR through careful grounding and filter stage redesign
- Collaborated with mechanical team on shield can and cable strain relief

University Senior Design — Smart Motor Controller

Team Electrical Lead | 2018

Capstone project delivering a complete motor control demo with safety interlocks.

- Owned power stage schematic and current sensing accuracy budget
- Presented final demo to faculty and industry judges

9. Professional Experience (Detailed)

Electrical Engineer — RedRiver Controls

Feb 2021 – Present | Austin, TX

Design and validate electronics for industrial control products. Own schematics, layout reviews, and lab characterization for power and sensing boards.

- Released two board revisions to production with full test coverage documentation
- Reduced field return rate related to connector intermittent faults by redesigning strain and ESD path
- Led schematic reviews attended by firmware, mechanical, and quality stakeholders
- Built automated Python scripts to log electronic-load sweep data for regression comparisons
- Supported contract manufacturer with first-article electrical measurements and yield triage

Junior Electrical Engineer — OrbitSense IoT

Aug 2018 – Jan 2021 | Austin, TX

Worked on battery-powered IoT sensor nodes: power budget, charging path, and RF-adjacent layout constraints.

- Designed Li-ion charging and protection circuit for a wearable-adjacent form factor
- Improved sleep current by ~30% through rail sequencing and firmware coordination
- Performed bring-up of ESP32-based prototypes and wrote flash/test station checklist
- Assisted EMI pre-compliance; added filtering that passed subsequent lab visit

Electrical Engineering Intern — Texas Power Labs

Summer 2017 | Round Rock, TX

Supported lab characterization of power supply reference designs.

- Automated repetitive efficiency measurements and plotted results for senior engineers
- Learned disciplined lab safety and instrument care practices

10. Extended Experience Highlights

I approach electrical design as a system problem: power integrity, signal integrity, thermal behavior, firmware timing, and manufacturability must be considered early. I document assumptions so future spins are cheaper and safer.

Additional highlights:

- Created a reusable first-power-on checklist adopted by the hardware team
- Partnered with procurement on second-source ICs during a supply shortage
- Presented lunch-and-learn on return current paths and common EMI mistakes
- Supported field FA by reproducing customer load profiles on the bench
- Mentored a new graduate engineer on Altium libraries and schematic hygiene

11. Hobbies & Interests

My hobbies keep me curious about both electronics and the wider world.

- DIY audio amplifier and guitar pedal experiments (safe low-voltage projects)
- Amateur radio study / Morse practice sessions with local club friends
- Trail running and weekend camping in Texas Hill Country
- Reading popular science books on energy systems and semiconductors
- Mentoring university students at hardware hack nights

12. Closing Note

Thank you for considering my application. I would be glad to discuss how my electrical engineering experience in power, sensing, and embedded bring-up can support your product roadmap. References on request.